

Functional genomics reveals the mechanism of hypoxic adaptation in nontuberculous mycobacteria

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Abstract

Mycobacterium intracellulare is a major etiological agent of the recently expanding *Mycobacterium avium*–*intracellulare* complex pulmonary disease (MAC-PD). Therapeutic regimens that include a combination of macrolides and antituberculous drugs have been implemented with limited success. To identify novel targets for drug development that accommodate the genomic diversity of *M. avium*–*intracellulare*, we subjected eight clinical MAC-PD isolates and the type strain ATCC13950 to genome-wide profiling to comprehensively identify universally essential functions. Among these strains, we identified 131 shared essential or growth-defect-associated genes. Unlike the type strain, the clinical strains showed an increased requirement for genes involved in gluconeogenesis and the type VII secretion system under standard growth conditions, the same genes required for hypoxic pellicle-type biofilm formation in ATCC13950. Consistent with the central role of hypoxia in the evolution of *M. intracellulare*, the clinical MAC-PD strains showed more rapid adaptation to hypoxic growth than the type strain. Importantly, the increased essentiality of hypoxic fitness genes was confirmed in a mouse lung infection model. These findings confirm the concordant gene essentiality under hypoxic conditions *in vitro* and hypoxia-related conditions *in vivo*, and highlight the importance of using clinical strains and host-relevant growth conditions to identify high-value targets for drug development.

eLife assessment

This descriptive study reports the genetic requirements for growth and fitness of multiple clinical strains of a relatively understudied species of mycobacteria, *Mycobacterium intracellulare*. The findings are **valuable** however, the study is **incomplete** as the primary claims related to hypoxia adaptation need additional experimental support and data presentation requires more clarity. The work will be of interest to microbiologists.

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Introduction

In contrast to the ongoing decline in the rate of tuberculosis, *Mycobacterium avium–intracellulare* complex pulmonary disease (MAC-PD) is increasing in many parts of the world^{1,2}. The standard therapy for MAC-PD involves multidrug chemotherapy that includes clarithromycin and several antituberculous drugs. Unfortunately, treatment failure is common, resulting in relapse, disease progression, and death³. Therefore, it is critical to investigate the bacterial physiological characteristics that underlie the ability of MAC-PD strains to establish and maintain chronic drug-recalcitrant infections.

Recent advances in comparative genomics have revealed major differences in the genomic features of reference type strains and circulating pathogenic clinical isolates. Extensive genomic diversity has been recognized across clinical isolates of nontuberculous mycobacteria (NTM). *Mycobacterium avium* has been divided into four subspecies including subspecies, *avium*, *hominissuis*, *silvaticum*, and *paratuberculosis*⁴. Whereas *M. intracellulare* has been divided into two subgroups, typical *M. intracellulare* (TMI) and *M. paraintracellulare–M. indicus pranii* (MP-MIP), with currently two additional distinguished subspecies, *yongonense*, and *chimaera*^{5,6}. Despite the increasing number of sequenced MAC-PD strains, the molecular genetic mechanisms that underlie their virulence and pathogenesis are still poorly understood.

The burgeoning field of functional genomics has allowed the rapid assessment of molecular aspects of organisms on a genome-wide scale. Transposon (Tn) sequencing (TnSeq) is a functional genomic approach that combines saturation-level transposon-insertion mutagenesis with next-generation sequencing (NGS) to comprehensively evaluate the fitness costs associated with gene inactivation in bacteria. Over the last decade, TnSeq has been applied to numerous bacterial species using various approaches⁷. Whereas initial studies applied TnSeq to reference strains, recent approaches have included the comparative assessment of representative clinical strains to better understand the fundamental aspects of pathogen biology⁸. We have previously used TnSeq to globally characterize the genes that are essential for growth versus those that are essential for hypoxic pellicle formation in the *M. intracellulare* type strain ATCC13950⁹. However, given the recent expansion of MAC-PD and the well-recognized high levels of genomic diversity in this complex of bacteria, data obtained from the study of a decades-old reference strain is not expected to represent the diversity of extant clinical isolates. ATCC13950 was originally isolated from the abdominal lymph node of the 34-month-old female¹⁰. Even though the etiological bacterial species are the same, the clinical manifestation of infant lymphadenitis differs markedly from MAC-PD, which often occurs in middle-aged and elderly female patients with no predisposing immunological disorder. Therefore, it is essential to investigate a collection of strains that captures the diversity of recent clinical MAC-PD strains in order to define key molecular aspects of their pathogenesis and virulence.

In this study, we used TnSeq to analyze a set of recently isolated clinical *M. intracellulare* strains with diverse genotypes. We integrated the gene essentiality data for these strains and the type strain to identify the common essential and growth-defect-associated genes that represent the genomic diversity of this group of pathogens. We also identified the greater essentiality of genes involved in gluconeogenesis, the type VII secretion system, and cysteine desulfurase in the clinical MAC-PD strains than in the type strain. Furthermore, the requirement for these genes was confirmed in a mouse lung infection model. The profiles of gene essentiality in clinical MAC-PD strains suggest the mechanism of hypoxic adaptation in NTM in terms of promising drug targets, especially for strains that cause clinical MAC-PD.

Results

Common essential and growth-defect-associated genes representing the genomic diversity of *M. intracellulare* strains

To obtain comprehensive information on the genome-wide gene essentiality of *M. intracellulare*, we included nine representative strains of *M. intracellulare* with diverse genotypes in this study (Fig. 1a, b). We used TRANSIT¹¹ to obtain 2–6 million reads of Tn insertions. The average number of Tn insertion reads was > 50 at Tn-inserted TA sites, which confirmed the TnSeq data (Supplementary Table 1). The numbers of essential genes in the clinical MAC-PD strains, calculated with the hidden Markov model (HMM; a transition probability model), tended to be smaller than that in ATCC13950 (Supplementary Table 2). In total, 131 genes were identified as essential or growth-defect-associated with the HMM analysis across all *M. intracellulare* strains, including the eight MAC-PD clinical strains and ATCC13950 (Fig. 2a, Supplementary Table 3). The genes identified as universal essential or growth-defect-associated were involved in fundamental functions, such as glyoxylate metabolism, purine/pyrimidine metabolism, amino acid synthesis, lipid biosynthesis, arabinogalactan/peptidoglycan biosynthesis, porphyrin metabolism, DNA replication, transcription, amino acid tRNA ligases, ribosomal proteins, general secretion proteins, components of iron–sulfur cluster assembly, some ABC transporters, and the type VII secretion system. The genes identified corresponded to the genes that encode conventional and under-development drug targets, such as those encoding gyrase, arabinosyltransferase, enoyl-(acyl-carrier-protein) reductase, alanine racemase, DNA-directed RNA polymerase, malate synthase¹² and F₀F₁ ATP synthase (Fig. 2b). These data suggest that the set of universal essential or growth-defect-associated genes identified with the HMM analysis of *M. intracellulare* strains with diverse genotypes has potential utility as antibiotic drug targets.

By comparative genomic analysis, we have revealed 3,153 core genes among 55 *M. intracellulare* strains including those used in this study⁵. Of these core genes, 439 genes were classified as strain-dependent essential or growth-defect-associated with the HMM analysis (Supplementary Table 4). And among 5,824 accessory genes, 222 genes were classified as essential or growth-defect-associated with the HMM analysis. The proportion of these strain-dependent and accessory essential or growth-defect-associated genes was comparable to the similar analysis of *Streptococcus pneumoniae*¹³. Of note, 17 (3.8%) of strain-dependent essential or growth-defect-associated genes and 14 (6.3%) of accessory essential or growth-defect-associated genes were also included in the 175 genes required for hypoxic pellicle formation in ATCC13950 reported in our previous study⁹ (Supplementary Table 4).

Genes with increased gene essentiality in clinical MAC-PD strains are also required for hypoxic pellicle formation in the type strain

The sharing of strain-dependent and accessory essential or growth-defect-associated genes with genes required for hypoxic pellicle formation in ATCC13950 prompts us to consider the possibility that the clinical MAC-PD strains increase the gene essentiality of the genes involved in hypoxic

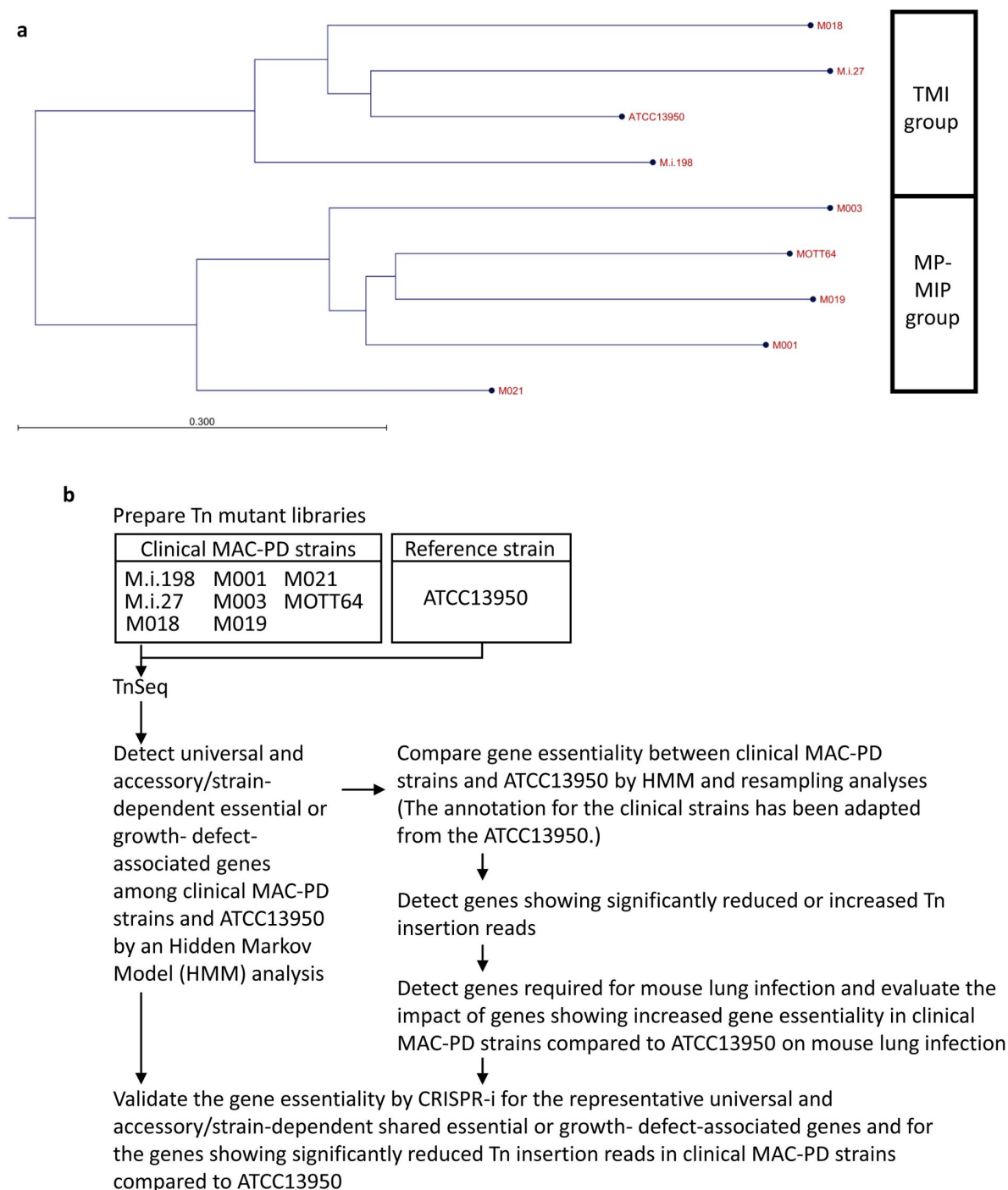


Fig. 1

Phylogenetic tree of the *M. intracellulare* strains and strategy of the experiment in this study.

a Phylogenetic tree of the *M. intracellulare* strains used in this study. The tree was generated based on average nucleotide identity (ANI) with the neighbor-joining method. TMI: typical *M. intracellulare*; MP-MIP: *M. paraintracellulare*-*M. indicus pranii*.
b Strategy of the experimental procedure.

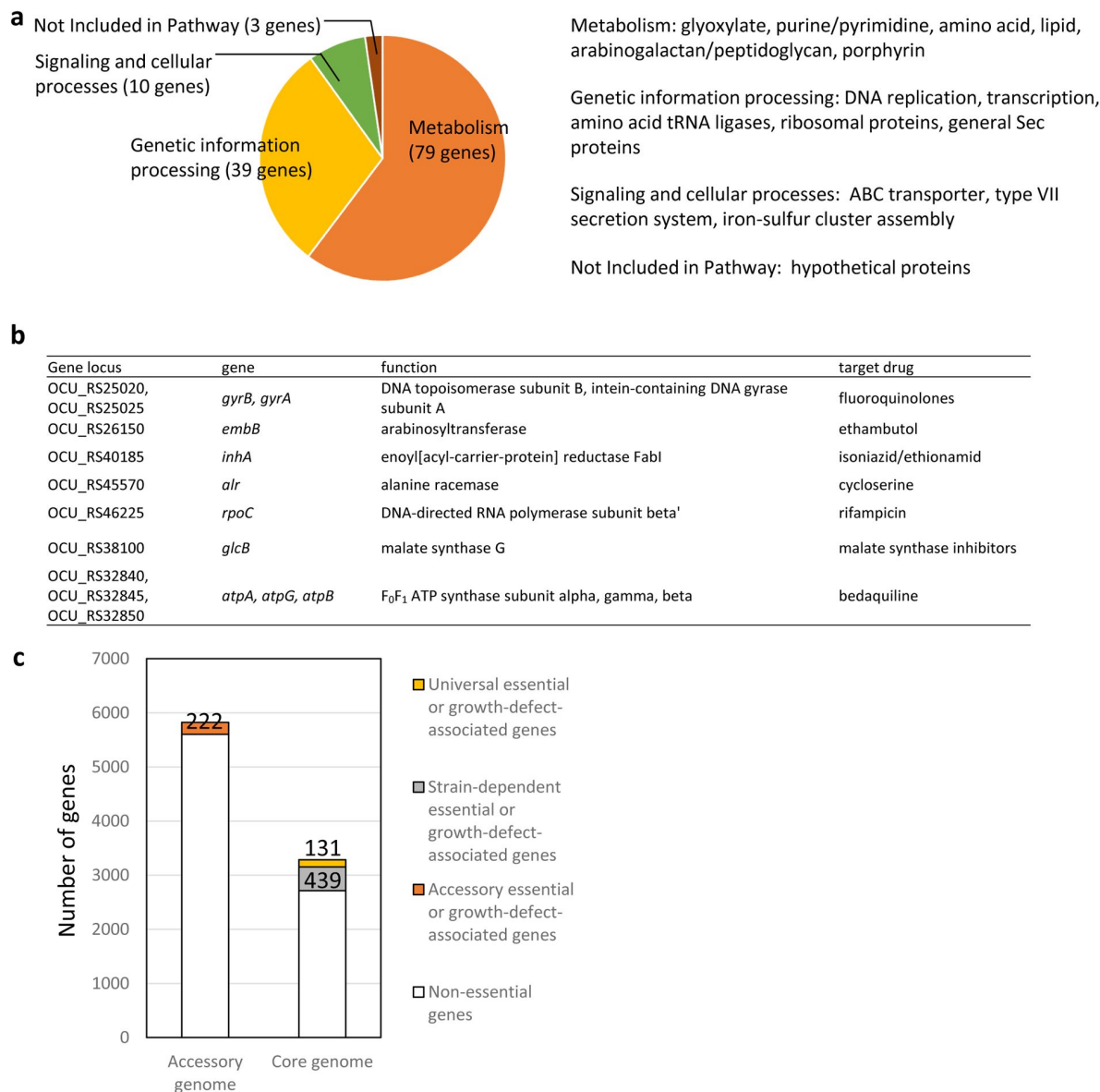


Fig. 2

Identification of the essential and growth-defect-associated genes across all nine *M. intracellulare* strains used in this study.

a Functional categories of 131 genes identified as universal essential or growth-defect-associated with an HMM analysis. The genes were categorized according to the Kyoto Encyclopedia of Genes and Genomes (KEGG) database. **b** Universal essential or growth-defect-associated genes corresponding to the genes of existing antituberculous drug targets. **c** The number of the essential or growth-defect-associated genes in the accessory genome, and the number of strain-dependent essential or growth-defect-associated genes in the core genome.

growth compared to ATCC13950. To clarify this, we compared the gene essentiality of the clinical MAC-PD strains with that in ATCC13950 by integrating the HMM analysis with the resampling analysis, a method of calculating the number of significantly altered Tn insertion reads (a gene-based permutation model). With this integrating analysis, 121 genes in the clinical MAC-PD strains were shown to have significantly fewer Tn insertions than ATCC13950 (**Fig. 3**, Supplementary Tables 5, 6). Of these, nine genes were identified in 3–5 clinical MAC-PD strains. These included genes encoding phosphatases (OCU_RS30185, OCU_RS35795), cysteine desulfurase (*csd* [OCU_RS40305]), methyl transferase (OCU_RS41540), a hypothetical protein (OCU_RS47290), type VII secretion components (*mycP5* [OCU_RS38275], *eccC5* [OCU_RS38345]), a glycine cleavage system aminomethyltransferase (*gcvT* [OCU_RS35955]), and phosphoenolpyruvate carboxykinase (*pckA* [OCU_RS48660]) (**Figs 3**, **4**). The genes involved in gluconeogenesis (such as *pckA*), the glycine cleavage system, and the type VII secretion system were also identified as required for hypoxic pellicle formation in ATCC13950 in our previous study⁹. The Tn insertion reads were significantly reduced in the pathways of gluconeogenesis, including in fructose-1,6-bisphosphatase (*glpX*), phosphoenolpyruvate carboxykinase (*pckA*), and pyruvate carboxylase (*pca*), but were significantly increased in those of glycolysis, including in pyruvate kinase (*pykA*) and pyruvate dehydrogenase (*aceE*, *lpdC*), suggesting that carbohydrate metabolism was metabolically remodeled in the clinical MAC-PD strains, which mimics the metabolism observed during hypoxic pellicle formation in ATCC13950 (**Fig. 5**).

On the other hand, 162 genes were detected with significantly more Tn insertions in the clinical MAC-PD strains than in ATCC13950 (**Fig. 3**). Of these, 12 genes were identified in more than seven strains. The increased Tn insertion reads in the genes encoding superoxide dismutase (OCU_RS25945) and pyruvate dehydrogenase (*aceE* [OCU_RS35720]) in the clinical MAC-PD strains suggest the inessentiality of the detoxification of oxygen radicals and glycolysis metabolism in these strains (**Figs 3–5**). These genes also included those encoding amidohydrolases (OCU_RS34400, OCU_RS34425, OCU_RS34445), some transcription factors (OCU_RS28440, OCU_RS30230), a CoA transferase (OCU_RS34390), and some predicted transporters (OCU_RS28435, OCU_RS39950, OCU_RS39955, OCU_RS44685). Furthermore, the increase in Tn insertion reads in predicted multidrug-resistant genes, such as those encoding the DrrAB-ABC transporter complex (OCU_RS39950, OCU_RS39955)¹⁴ and a putative lipoprotein LpqB (OCU_RS44685), encoded by the two-component system *mtrAB* operon^{15–16}, in the clinical MAC-PD strains were found.

Requirement of genes with increased gene essentiality in the clinical MAC-PD strains for mouse lung infection

Hypoxia is the major environmental condition in natural water¹⁷, biofilms¹⁸, and infections in humans and animals characterized by caseous granuloma lesions¹⁹. The impact of hypoxia on mycobacteria under various ecological circumstances implies that the genes with increased essentiality in the clinical MAC-PD strains compared to ATCC13950 and the genes required for hypoxic pellicle formation in ATCC13950 may be required for infection *in vivo*. To examine whether the set of genes with increased essentiality in the clinical MAC-PD strains also shows increased essentiality during infection *in vivo*, we performed TnSeq in mice infected intratracheally with the Tn mutant library strains of M.i.198 or M.i.27, which we had demonstrated to be virulent in mice (Supplementary Fig. 1a). The time course of the changes in the bacterial burden showed a pattern similar to those of the wild-type strains M.i.198 and M.i.27, respectively, except that it was not possible to harvest sufficient colonies (as few as 10⁴/mouse) in the few mice infected with the M.i.27 Tn mutant strain in week 8 and week 16 (Supplementary Fig. 1b, Supplementary Table 7). With a resampling analysis, 629 and 512 genes were shown to be required for *in vivo* infection with transposon mutant library strains M.i.27 and M.i.198, respectively, between day 1 and week 16 of infection (**Fig. 6a**, Supplementary Fig. 1c, d, Supplementary Tables 8, 9). Of these genes, 72 (11.9%) and 50 (9.76%) in M.i.27 and M.i.198, respectively, were also required for hypoxic pellicle formation in ATCC13950. Forty-one genes also required for hypoxic pellicle formation in ATCC13950 were identified in both M.i.27 and M.i.198,

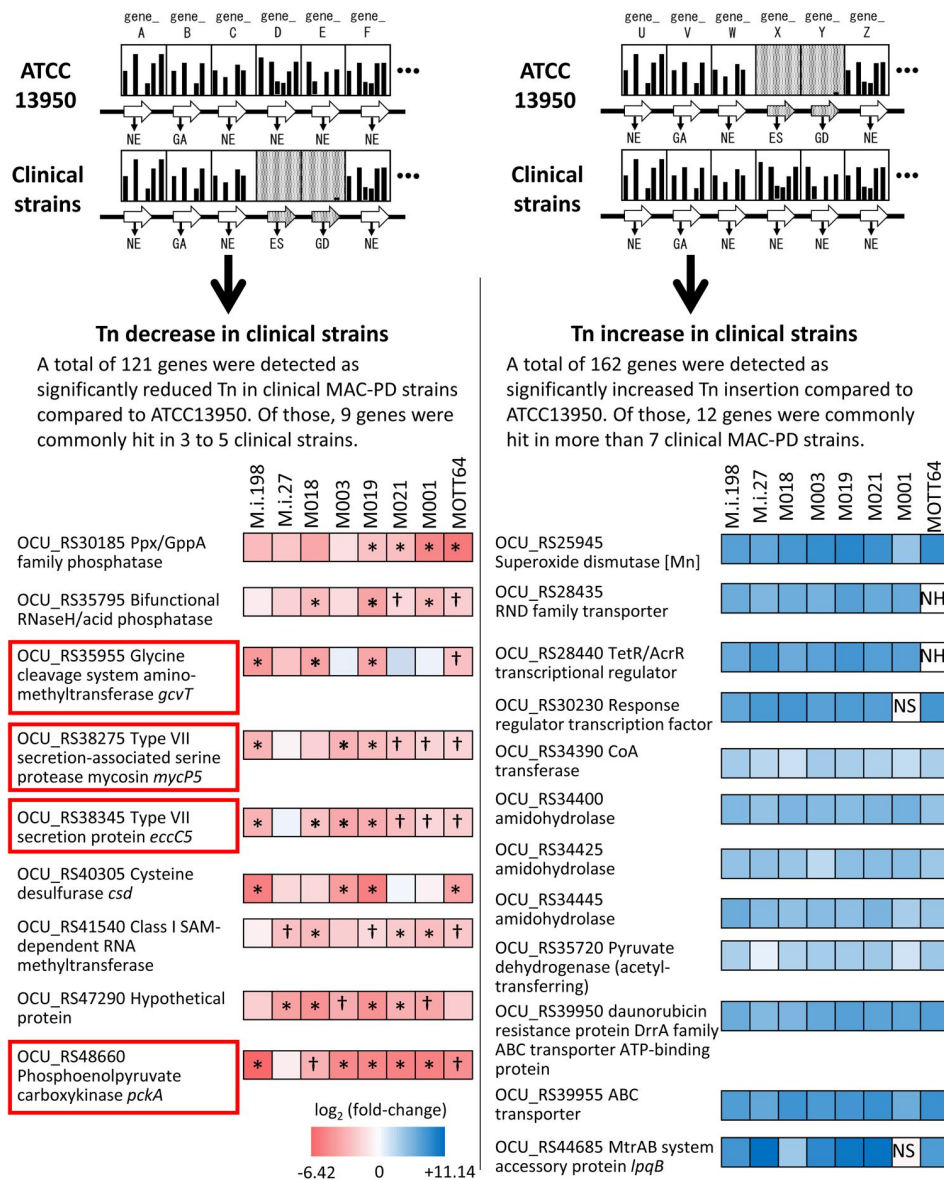


Fig. 3

Detection of genes showing increased or reduced gene essentiality in the clinical *M. intracellulare* strains.

Left panel shows the genes identified as having fewer Tn insertion reads than the type strain ATCC13950. The fold changes in the number of Tn insertion reads calculated by a resampling analysis are represented by the color scale. Red squares indicate genes required for hypoxic pellicle formation in ATCC13950^{30,31}. *Genes identified with a combination of resampling and HMM analyses. †Genes identified only with resampling analysis. Right panel shows the genes identified as having more Tn insertion reads than type strain ATCC13950. NH: no homolog with ATCC13950, NS: no significant increase in Tn insertion reads by a resampling analysis (adjusted $P < 0.05$).

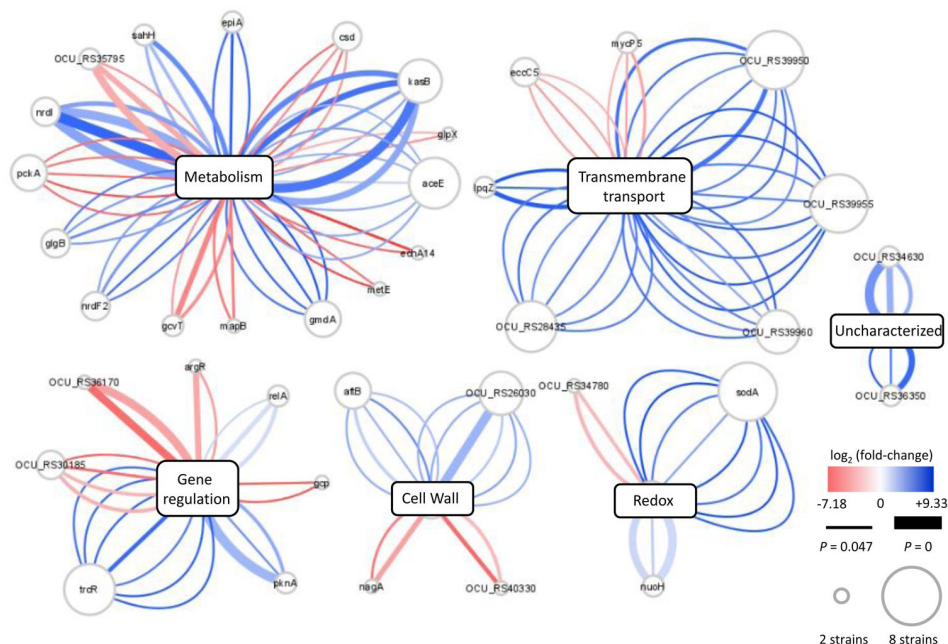


Fig. 4

Overview of the differences in gene essentiality between the clinical *M. intracellulare* strains and ATCC13950, drawn with Cytoscape⁵⁵.

Each central node represents the functional category of genes, assigned with a KEGG pathway analysis. Each peripheral node represents the genes showing significant changes in the number of Tn insertion reads. The size of each node represents the number of strains identified. Each edge represents a strain identified as showing significant changes in the number of Tn insertion reads. The thickness of each edge represents the adjusted *P* value. The color scale of each edge represents the fold change in the number of Tn insertion reads calculated by a resampling analysis. The graphical organization was referenced to the previous publication by Carey, *et al.*⁵⁶

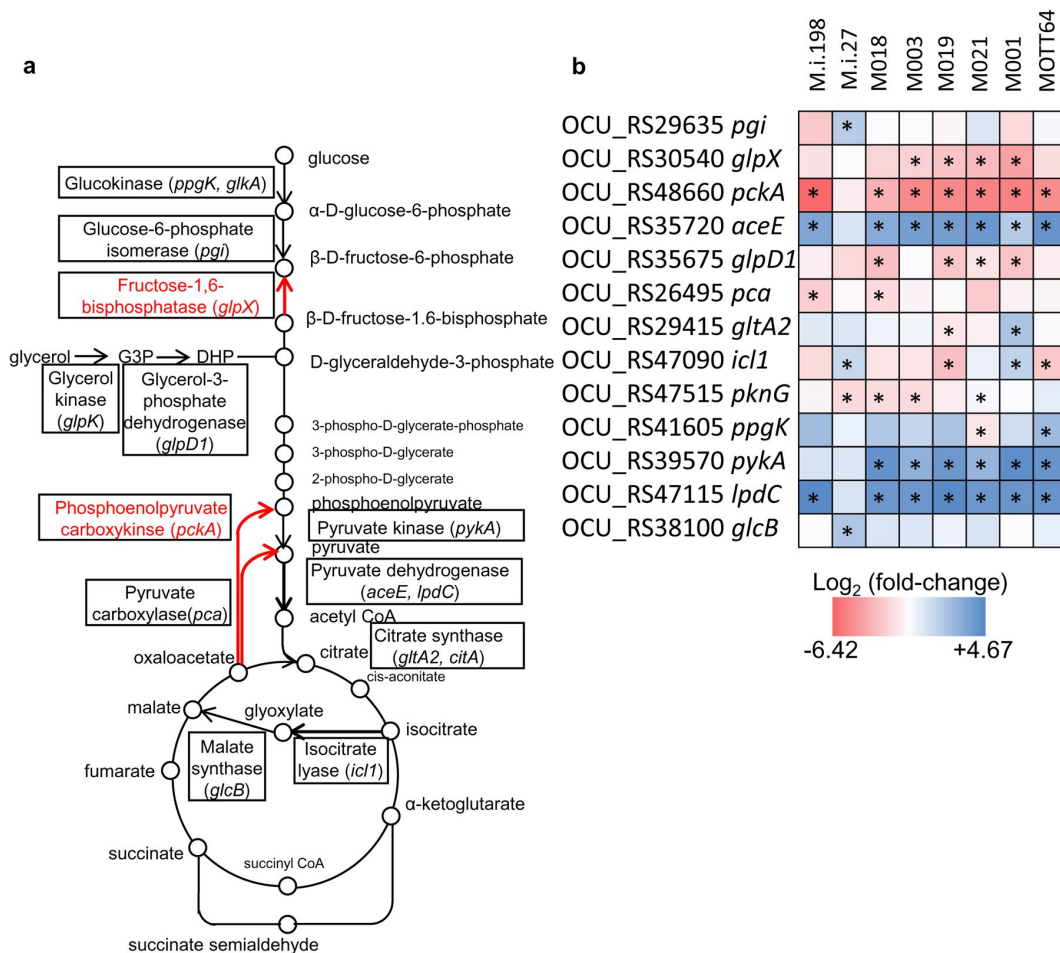


Fig. 5

Preferential gene requirements for gluconeogenesis in the clinical *M. intracellulare* strains inferred from the TnSeq results.

a Carbohydrate pathway showing the changes in Tn insertion reads. Genes identified with the resampling analysis are framed by squares. The essentiality of gluconeogenesis-related genes (*pckA*, *glpX*) was higher and that of glycolysis-related genes (*aceE*, *lpdC*) was lower in the clinical *M. intracellulare* strains compared with those in ATCC13950. **b** Data on gene essentiality calculated with a resampling analysis in the clinical *M. intracellulare* strains compared with ATCC13950. Red indicates a reduction in Tn insertion reads in the clinical *M. intracellulare* strains. Blue indicates an increase of Tn insertion reads in the clinical *M. intracellulare* strains. Asterisks indicate statistical significance of log₂ fold changes (log₂FCs). The graphical organization was referenced to the previous publication by Carey, *et al.*⁸

and included genes associated with succinate production (multifunctional 2-oxoglutarate metabolism enzyme) and the glyoxylate cycle (isocitrate lyase), cysteine and methionine synthesis (thiosulfate sulfurtransferase [CysA], 5-methyltetrahydrofolate-homocysteine methyltransferase [MetH]), the serine/threonine-protein kinase PknG system, proteasome subunits (PrcA, PrcB), DNA repair UvrC, secretion protein EspR, cell-wall synthesis proteins (such as phosphatidylinositol [polyphosphatidyl-mannose-dependent alpha-[1-2]-phosphatidylinositol pentamannoside mannosyltransferase], peptidoglycan (penicillin-binding protein A pbpA), and L,D-transpeptidase 2), putative membrane proteins and transporters (such as the MmpL family, transport accessory protein MmpS3, transport protein MmpL11, vitamin B₁₂ transport ATP-binding protein BacA, ABC transporter substrate-binding protein, and ABC transporter permease), and mammalian cell entry (Mce) proteins (OCU_RS48855–48875) (Supplementary Table 10).

The genes only identified in M.i.27 that were also required for hypoxic pellicle formation in ATCC13950 included genes associated with gluconeogenesis (*pckA*, *glpX*), the glycine cleavage system (*gcvT*), the type VII secretion system (*eccD5*, *espG5*, *eccC5* [OCU_RS38280-OCU_RS38285-OCU_RS38345], *eccC3* [OCU_RS48145]), and D-inositol 3-phosphate glycosyltransferase (Supplementary Table 10). These three systems, gluconeogenesis, the glycine cleavage system, and the type VII secretion system, also showed increased gene essentiality in 3–5 clinical MAC-PD strains than in ATCC13950 (Figs 3, 6). Although M.i.27 was excluded as a relevant strain showing increased gene essentiality of these three metabolic systems, the increase in gene requirements for infection *in vivo* suggests a positive relationship between hypoxic growth *in vitro* and bacterial growth *in vivo*. In contrast, M.i.198 already showed increased gene essentiality in these three metabolic systems under aerobic conditions *in vitro*, so the genes associated with these three systems were not identified as having fewer Tn insertion reads during infection *in vivo*. Genes only identified in M.i.198 and required for hypoxic pellicle formation in ATCC13950 included genes associated with succinate synthesis (2-oxoglutarate oxidoreductase subunit KorA) or encoding the HTH-type transcriptional repressor KstR and menaquinone reductase (Supplementary Table 10). As increased gene essentiality in 3–5 clinical strains (excluding M.i.198) compared with ATCC13950, an increased requirement for the RNA methyltransferase OCU_RS41540 was identified in M.i.198 during infection *in vivo*. Despite the differences in the profiles of the genes required for *in vivo* infection between strains, these data suggest that increased gene essentiality for hypoxic growth confers advantages for pathogenesis *in vivo*.

We have performed a statistical enrichment analysis of gene sets by GSEA²⁰ to investigate whether the genes required for growth of M.i.27 and M.i.198 in mouse lungs overlap the gene sets required for hypoxic pellicle formation in ATCC13950 as well as the gene sets showing increased gene essentiality observed in the clinical MAC-PD strains listed in Fig 3. Of these gene sets including a total of 180 genes, 54% (98 genes) and 40% (73 genes) were listed as enriched in genes required for mouse infection in M.i.27 and M.i.198, respectively ($P < 0.01$ in M.i.27 and $P = 0.016$ in M.i.198) (Supplementary Fig. 2, Supplementary Table 11). The enriched genes both in M.i.27 and M.i.198 included genes encoding fructose 1,6-bisphosphatase (*glpX*), a type VII secretion protein (*eccC5*), glycine cleavage system aminomethyltransferase (*gcvT*), some phosphatase (OCU_RS30185) and a hypothetical protein (OCU_RS47290) (Fig. 6). The enriched genes only in M.i.27 included genes of phosphoenolpyruvate carboxylase (*pckA*), cysteine desulfurase (*csd*) and the ESX-5 type VII secretion component serine protease (*mycP5*). The enriched genes only in M.i.198 included genes encoding some phosphatase (OCU_RS35975) and class I SAM-dependent RNA methyltransferase (OCU_RS41540). These data suggest the sharing of the hypoxia-adaptation pathways between *in vitro* hypoxic pellicle formation and *in vivo* bacterial growth.

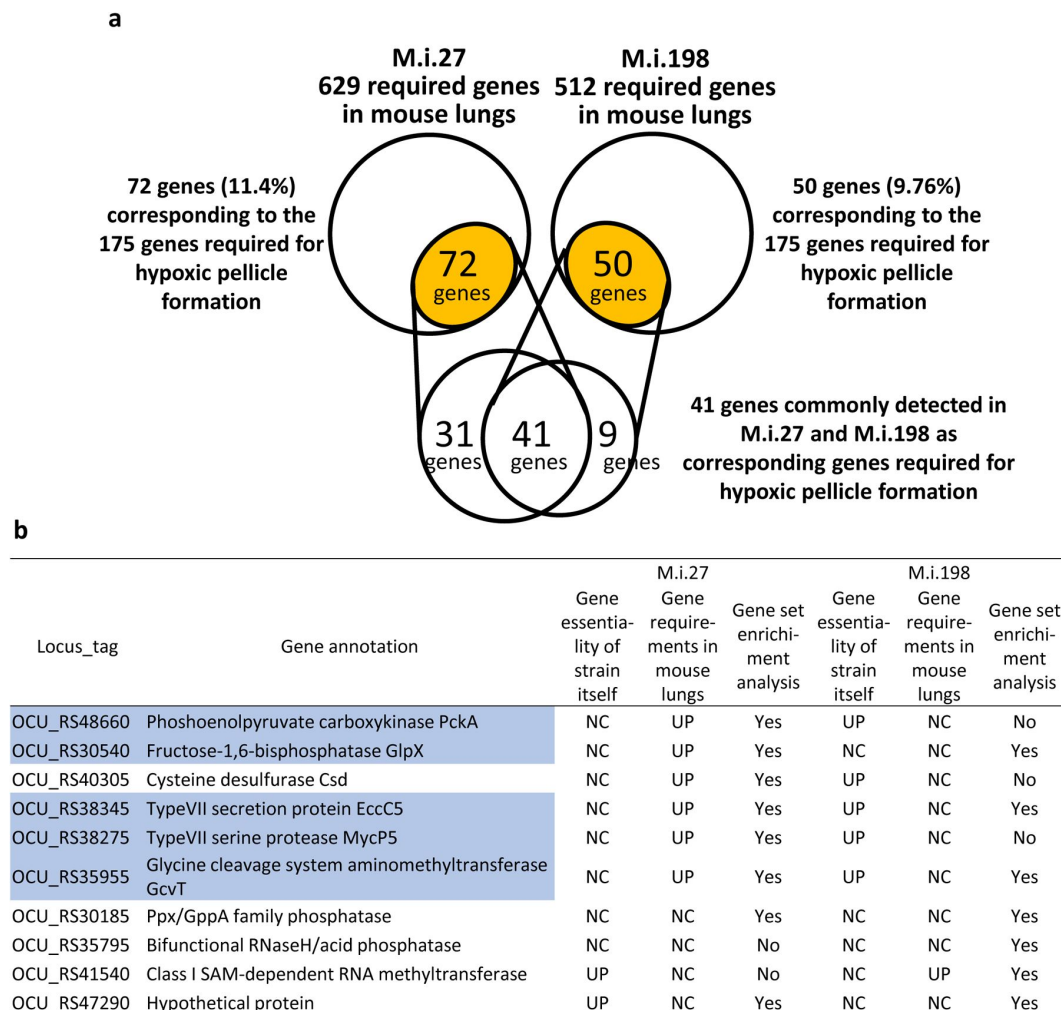


Fig. 6

Significance of increased gene essentiality profiles *in vitro* on the pathogenesis of mouse lung infection.

a Consistency between the genes required for infection in mouse lungs and the 175 genes required for hypoxic pellicle formation in ATCC13950. **b** Changes in the requirements for genes with increased essentiality observed in the clinical *M. intracellulare* strains after the infection of mouse lungs. Highlighted genes are also required for hypoxic pellicle formation by ATCC13950. Gene essentiality of the strain itself: gene essentiality compared with that of ATCC13950. Gene requirements in mouse lungs: changes in the gene requirements in the infected mouse lungs compared with those before infection. Gene set enrichment analysis: The genes listed as core enrichment are shown as “Yes” and the genes not listed as core enrichment are shown as “No.”

Preferential hypoxic adaptation of clinical MAC-PD strains evaluated with bacterial growth kinetics

The metabolic remodeling, such as the increased gene essentiality of gluconeogenesis and the type VII secretion system, and the overlap of the genes required for mouse lung infection and those required for hypoxic pellicle formation involved by confer these metabolic pathways suggests the preferential adaptation of the clinical MAC-PD strains to hypoxic conditions. To confirm this, we investigated the growth kinetics of each strain under aerobic and hypoxic conditions. In terms of growth kinetics, most of the clinical MAC-PD strains (except M001) entered logarithmic (log) phase earlier than ATCC13950 under 5% oxygen (**Fig. 7a** [↗](#), Supplementary Fig. 3). However, under aerobic conditions, the clinical MAC-PD strains entered log phase at a similar time to ATCC13950. The growth rate at mid-log point (midpoint) was significantly reduced under hypoxic conditions in a total of 5 clinical MAC-PD strains (M.i.198, M.i.27, M003, M019 and M021) compared to aerobic conditions (**Fig. 7b** [↗](#), Supplementary Fig. 3). The growth rate at midpoint under hypoxic conditions was significantly lower in these 5 clinical MAC-PD strains than in ATCC13950. By contrast, the growth rate at midpoint was significantly increased under hypoxic conditions compared to aerobic conditions in ATCC13950. And there was no significant change in the growth rate between aerobic and hypoxic conditions in the rest 3 clinical MAC-PD strains (M018, M001 and MOTT64). The data of more rapid adaptation to hypoxia followed by reduced growth rate once adapted indicate the greater metabolic advantage on continuous logarithmic bacterial growth under hypoxic conditions in the clinical MAC-PD strains than in ATCC13950.

Effects of knockdown of universal and accessory/strain-dependent essential or growth-defect-associate genes in clinical MAC-PD strains

To facilitate the functional analysis of the TnSeq-hit genes, it is necessary to construct the genetically-engineered strains. Still now, there are very few reports of gene engineering of MAC strains and none of them has been accompanied by the follow-up studies^{21 [↗](#)–26 [↗](#)}. Moreover, no attempts have been reported to construct the knockdown strains in MAC strains. With an intention to break through this current situation, we used the pRH2052/pRH2521 clustered regularly interspaced short palindromic repeats interference (CRISPR-i) system to construct knockdown strains where the expression of the target gene is suppressed^{27 [↗](#)}. Of the nine *M. intracellulare* strains analyzed in this study, seven could be transformed with the vectors for single guide RNA (sgRNA) expression and dCas9 expression. However, the other two strains, M001 and MOTT64, could not be transformed with these CRISPR-i vectors. To confirm the gene essentiality detected with the HMM analysis, we evaluated the consequent growth inhibition in the knockdown strains of representative universal essential or growth-defect-associated genes, including *glcB*, *inhA*, *gyrB*, and *embB*. The knockdown strains showed growth suppression with comparative growth rates of knockdown strains to vector control strains that do not express single sgRNA reaching 10^{-1} to 10^{-4} (**Fig. 8a** [↗](#)).

As representative accessory and strain-dependent essential or growth-defect-associated genes, we evaluated the 9 genes showing reduced Tn insertion reads in the 3–5 clinical MAC-PD strains as listed in **Fig. 3** [↗](#), as well as *glpX*, a gene of gluconeogenesis where four MAC-PD strains showed reduced Tn insertion reads in a resampling analysis (**Fig. 5b** [↗](#)). The growth was suppressed in the *pckA*-knockdown strains in M.i.198 and M003, in the *glpX*-knockdown strain in M003, and in the cysteine desulfurase gene (*csd*)-knockdown strains in M.i.198 and M003, with comparative growth rates of knockdown strains to vector control strains reaching 10^{-2} to 10^{-4} (**Fig. 8b** [↗](#)). The growth tended to be suppressed in the type VII secretion protein gene (*eccC5*)-knockdown strain in M003 and in the type VII secretion-associated serine protease mycosin gene (*mycP5*)-knockdown strain in M003, with comparative growth rates of knockdown strains to vector control strains reaching

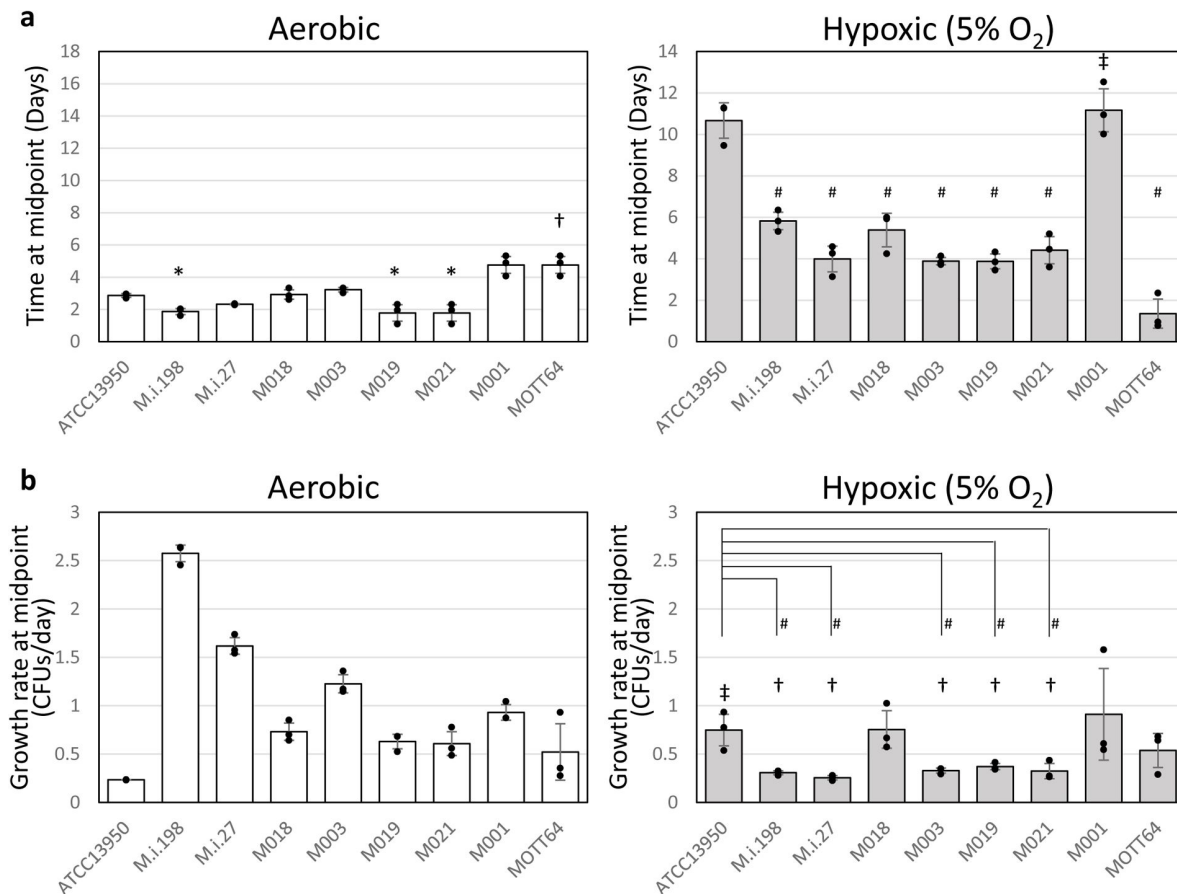


Fig. 7

Comparison of the timing of entry into logarithmic growth and logarithmic growth rate by the clinical *M. intracellulare* strains and ATCC13950.

a Time at the inflection point (midpoint) on the sigmoid growth curve. *significantly earlier than an aerobic culture of ATCC13950; #significantly earlier than a hypoxic culture of ATCC13950; †significantly later than an aerobic culture of ATCC13950; ‡significantly later than a hypoxic culture of ATCC13950. **b** Growth rate at midpoint of the growth curve in each strain. #significantly slower than a hypoxic culture of ATCC13950; †significantly slower than an aerobic culture of the corresponding strain; ‡significantly faster than an aerobic culture of ATCC13950. Open bars: aerobic; closed bars: 5% O₂. Data are shown as the means $\pm$ SD of triplicate experiments. Data from one experiment representative of three independent experiments ($N = 3$) are shown.

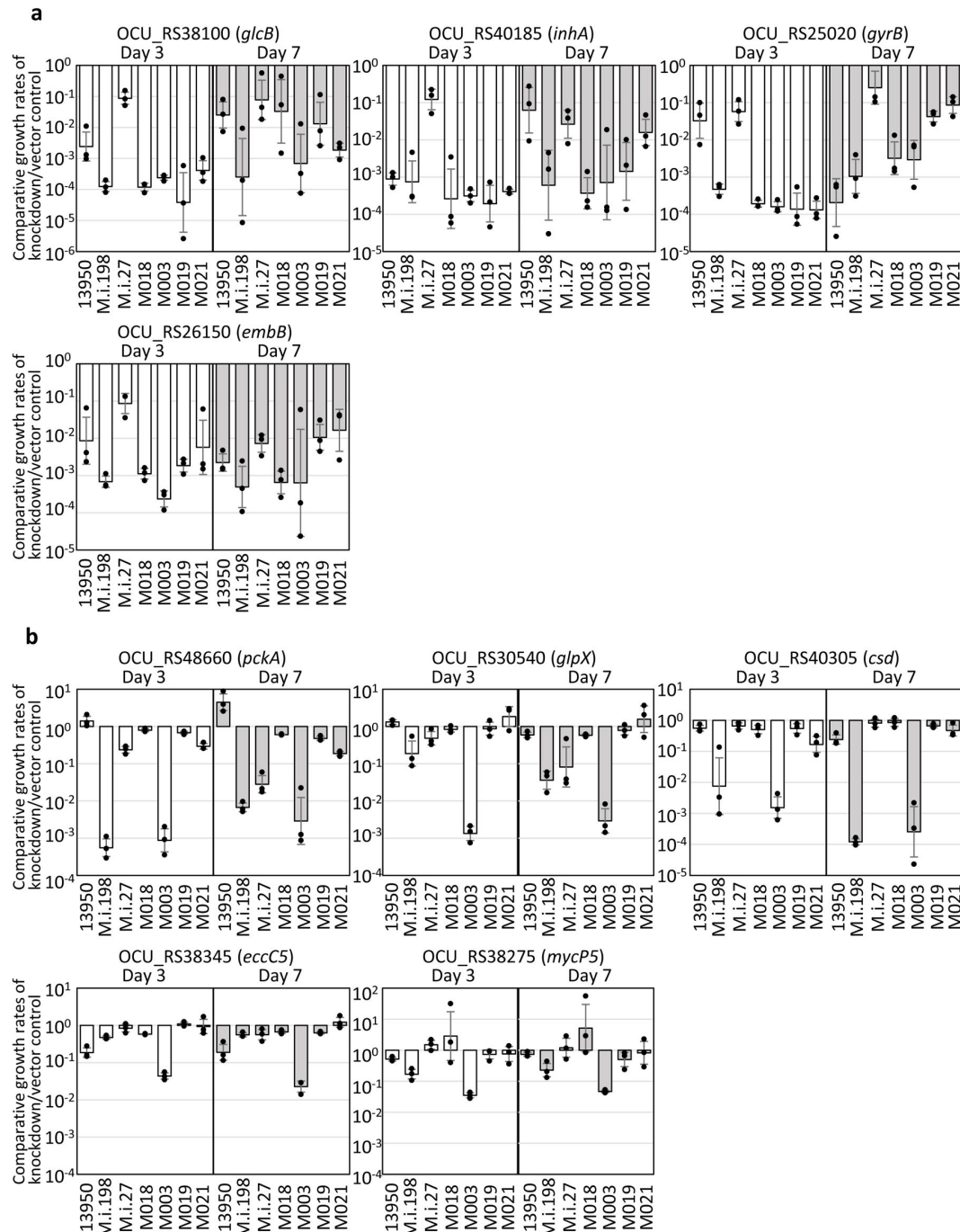


Fig. 8

Validation of the inhibition of bacterial growth caused by the suppression of the expression of TnSeq-hit genes using the CRISPR-i system.

Open bar: day 3; closed bar: day 7. **a** Comparative growth rates of the knockdown strains relative to those of the vector control strains in the representative universal essential or growth-defect-associated genes: *glcB*, *inhA*, *gyrB* and *embB*. **b** Comparative growth rates of the knockdown strains relative to those of the vector control strains in the representative accessory and strain-dependent essential or growth-defect-associated genes: *pckA*, *glpX*, *csd* and ESX-5 type VII secretion components. Data are shown as the means $\pm$ SD of triplicate experiments. Data from one experiment representative of two independent experiments ($N = 2$) are shown.

less than 10^{-1} . By contrast, the growth was not suppressed in these knockdown strains in ATCC13950, with comparative growth rates of knockdown strains to vector control strains not reaching 10^{-1} , which was consistent to the inessentiality of these genes under aerobic conditions in ATCC13950. As for M.i.198 and M003, the consistency between TnSeq data and CRISPR-i data was supported by a correlation between the growth suppression of the knockdown strains and the fitness costs evaluated by a resampling analysis (Supplementary Table 12).

In contrast to the cases of M.i.198 and M003, the growth of knockdown strains of the accessory and strain-dependent essential genes was not effectively suppressed in the other MAC-PD strains despite positively detected in TnSeq (**Fig. 5b**, Supplementary Fig. 4). These data suggest the difficulties in the effective operation of genetically modified system using foreign genes from different bacterial species in MAC-PD strains. Although the suppression of gene expression was confirmed in the knockdown strains of these accessory and strain-dependent essential or growth-defect-associate genes, the gene expression levels were not suppressed to $< 1\%$ as reported in *M. tuberculosis* (Mtb)²⁸ (Supplementary Fig. 5). The differences between strains as well as between genes in the degree of suppression of gene expression required to inhibit bacterial growth may explain such inconsistencies between the TnSeq and CRISPR-i data.

To be summarized, we could validate the universal essential or growth-defect-associated genes by the CRISPR-i system in the *M. intracellulare* strains used in this study. And, although not successful with all strains, we could validate the accessory and strain-dependent essential or growth-defect-associated genes, especially genes of gluconeogenesis, type VII secretion components and iron-sulfur cluster assembly in several MAC-PD strains like M.i.198 and M003.

Discussion

In this study, using TnSeq of nine genetically diverse *M. intracellulare* strains, we have identified 131 shared essential and growth-defect-associated genes with an HMM analysis. The genes identified are promising drug targets, representing the genomic diversity of the clinical MAC-PD strains and reflecting the actual clinical situation. The strategy of narrowing the essential genes of bacteria is a useful approach for refining the database of drug targets. We have also demonstrated an overview of the differences in gene essentiality required for hypoxic growth between the ATCC type strain and the clinical *M. intracellulare* strains responsible for MAC-PD, with the evidence of earlier entry to log phase followed by slower growth rate observed in most of these clinical strains compared to ATCC13950. This is the first study to demonstrate the positive relationship between the profiles of increased gene essentiality required for hypoxic metabolic remodeling and the phenotypes advantageous for bacterial survival under hypoxic conditions.

In this study, 121 genes were detected as significantly reduced Tn insertion in the clinical MAC-PD strains compared to ATCC13950, which was higher than the case of Mtb showing 10–50 genes with altered Tn insertion reads detected by a resampling analysis⁸. This may reflect the genomic diversity of *M. intracellulare* compared with that of Mtb. The proportion of accessory genes is larger in NTM than in Mtb²⁹. In a clinical trial of phage therapy for *M. abscessus* infection, the number of accessory genes was shown to decrease with the loss of virulence in *M. abscessus* during the course of therapy³⁰, suggesting some role for accessory genes for its clinical pathogenesis. Our data of the overlap of the accessory essential or growth-defect-associated genes such as *pckA* and *mycP5* (Supplementary Table 5) with genes required for mouse lung infection (**Fig. 6b**, Supplementary Fig. 2) supports the idea that accessory genes (non-core genes) may encode functions required for bacterial survival, especially under nonoptimal conditions, such as hypoxia and during the infection of human and animal bodies.

Our TnSeq data are consistent with metabolomic data that have shown the importance of the gluconeogenic carbon flow of TCA cycle intermediates for bacterial growth under hypoxic conditions in terms of the essentiality of PckA and the reverse TCA cycle^{31–34}. Gluconeogenesis is essential for bacterial growth when fatty acids are used as a major nutrient³³. PckA catalyzes the conversion of the metabolites derived from the TCA cycle to the metabolites of gluconeogenesis³². The reaction catalyzed by PckA is essential for bacterial growth when the available nutrients are predominantly fatty acids^{35,36}. The main nutrients for mycobacteria inside hypoxic granulomas in an infected human or mouse body are lipids, such as fatty acids and cholesterol, rather than carbohydrates^{33,37}. In the present study, *M. intracellulare* grew under 5% oxygen, whereas *Mtb* stops growing under 1% oxygen and enters a dormant state³². The depletion of phosphoenolpyruvate in the *pckA* deletion mutant of *Mtb* reduced the phosphoenolpyruvate–carbon flux essential for bacterial growth and drug sensitivity³². The present study supports the idea that PckA is essential for the microaerobic growth of mycobacteria. Our TnSeq data imply the preferential adaptation of the clinical MAC-PD strains for circumstances *in vivo*, which are characterized by hypoxia.

On the other hand, such metabolic shift towards gluconeogenesis and reverse TCA cycle can also be observed under reactive nitrogen specie and *in vivo* infection as well. This suggests that the metabolic shift observed under hypoxia is a general stress response rather than being specific to hypoxia due to the nature of the provided/available carbon source³⁸. Several factors other than hypoxia can be estimated to be related to the change of the gene essentiality in the clinical MAC-PD strains such as host-related factors and general stress conditions (i.e. nutrition of abundant fatty acids and less carbohydrates, low pH circumstances)^{33,35–37}. Except for our data of the profile of genes required for hypoxic pellicle formation in *M. intracellulare*⁹, no fundamental genome-wide data are available on the change of the gene essentiality in response to each stress factors in non-tuberculous mycobacteria. Further studies are needed to discriminate which factors are specifically involved in the increased essentiality of each gene. Although of interest, it is beyond the scope of the current study.

About 10% of the genes required for mouse infection are also required for hypoxic pellicle formation. The genes showing increased gene essentiality in the clinical MAC-PD strains, such as *pckA* and genes encoding the ESX-5 type VII secretion system, including *eccC5* and *mycP5*, also showed increased gene requirements in mouse infections. In a TnSeq study of mice infected intravenously with *M. avium*, the genes of the ESX-5 type VII secretion system were required for infection of the liver and spleen³⁹. The ESX-5 type VII secretion system is involved in the transport of fatty acids and the virulence of mycobacteria^{40,41}. These similarities in the profiles of gene essentiality prompt us to propose some role of biofilm formation in mycobacterial pathogenesis in animals and humans. In the present study, we shed light on the impact of hypoxia on biofilm formation, the acquisition of virulence, and the *in vivo* pathogenesis of NTM in terms of global gene essentiality. Although the morphologies of the pellicle and macroscopic lesions (including cavitation and nodular bronchiectasis) are not similar, monitoring the expression of the genes identified in this study may provide clues to the mechanism of biofilm formation *in vivo*.

In this study, we revealed the pattern of hypoxic adaptation characteristic to the clinical MAC-PD strains by the growth curve analysis: an early entry to log phase followed by a reduced growth rate (Fig. 7). This pattern implicates the continuous impact on the infected hosts during logarithmic bacterial growth, which may be involved in the persistent and steadily progressive illness for years in humans³. The advantage of slow growth in mycobacteria may be inferred by the role of a histone-like nucleoid protein, Mycobacterial DNA-binding protein 1 (MDP1), also designated HupB^{42–44}. The expression of MDP1 is enhanced in the stationary phase, resulting in reduced growth speed and long-term survival with increased resistance to reactive oxygen species (ROS)^{42–44}. The reduced growth rate after entry to log phase may allow long-term

survival of mycobacteria under ROS-rich circumstances inside host cells. The pattern of hypoxic adaptation as observed in the clinical MAC-PD strains may enhance the pathogenesis by extending the period of the repeated process of bacterial invasion and replication in infected macrophages.

Contrary to our expectation, not all clinical strains showed rapid adaptation from aerobic to hypoxic conditions relative to ATCC13950. These data suggest that some, but not a major proportion of, clinical *M. intracellulare* strains do not have an advantageous phenotype for continuous logarithmic growth under hypoxia. In *Mtb*, the adaptation to hypoxia is considered to be necessary for clinical pathogenesis because the conditions *in vivo*, such as inside macrophages or granulomas, are hypoxic^{38,45}. However, our strain M001 entered log phase later than ATCC13950 under 5% oxygen. Our inability to construct knockdown strains in M001 prevented us from clarifying its clinical pathogenesis without preferential hypoxic adaptation. However, because we have shown that most of the *M. intracellulare* strains with different genotypes showed preferential adaptation to hypoxia in this study, our results largely reflect the clinical situation of the etiological agents of MAC-PD. A possible strategy for identifying factors that confer clinical pathogenesis will be to focus on the major clinical strains that show preferential hypoxic adaptation.

The data of validation of the universal essential or growth-defect associated genes by the CRISPR-i system was overall acceptable in the clinically MAC-PD strains whose knockdown strains could be constructed (Fig. 8a). However, we have encountered the difficulties of validating the TnSeq-hit genes by CRISPR-i experiment, especially accessory and strain-dependent essential or growth-defect-associated genes in the MAC-PD strains other than M.i.198 or M003 (Fig. 8b, Supplementary Fig. 4). As a possible reason, the low efficiency of the CRISPR-i system may obscure the increase in gene essentiality suggested by the TnSeq data (Supplementary Fig. 5). The efficiency of the suppression of gene expression by CRISPR-i in accessory and strain-dependent essential or growth-defect-associated genes was lower in our *M. intracellulare* strains than in *Mtb*²⁸. In *Mtb*, the expression of *mmpL3*, a gene encoding a mycolic acid transporter, is reported to be suppressed to 6% under 10 ng/ml anhydrotetracycline (aTc) and by < 1% under 100 ng/ml aTc in the knockdown strains, compared to strains expressing scrambled single guide RNAs (sgRNA)²⁸. The gene expression levels were suppressed in our knockdown strains by around 10%–40% under 50–200 ng/ml aTc relative to the levels found in strains not expressing sgRNA (Supplementary Fig. 5). The growth inhibition effect is theoretically dependent upon the degree to which gene expression is suppressed. On the other hand, the inconsistent growth phenotypes of knockdown strains of accessory and strain-dependent essential or growth-defect-associated genes between MAC-PD strains implies some differences in the degree of suppression of gene expression required to inhibit bacterial growth between genes as well as between bacterial strains. Despite the issue of the lower efficiency of the CRISPR-I system in *M. intracellulare* than in *Mtb*, this study has opened the avenue for functional analysis of NTM that leads to the development of novel drugs that target universal and accessory/strain-dependent essential metabolic pathways that has significant impact on bacterial growth, especially by making use of M.i.198 and M003 as optimal NTM strains for gene manipulation.

As in the *glpX*-knockdown strains, *csd* knockdown caused growth suppression in several MAC-PD strains. This type II Csd is involved in the biosynthesis of the iron–sulfur clusters of the SUF machinery. Compared with the type I cysteine desulfurase IscS, the SUF machinery is active under oxidative-stress and iron-limiting conditions⁴⁶. Representatives of iron–sulfur cluster proteins include components of the electron-transport system, such as NADH dehydrogenase I (Nuo) and succinate dehydrogenase (Sdh), which are components of respiratory complexes I and II, respectively. Under hypoxic conditions, the non-proton-translocating type II NADH dehydrogenase Ndh is upregulated, and cytochrome *bd* oxidase (CydAB) and the proton-pumping type I NADH dehydrogenase (Nuo) are downregulated³⁸. In the reverse TCA cycle on the same reaction pathway as Sdh, fumarate reductase (Fnr) is also an iron–sulfur cluster protein that produces succinate, and is a proposed alternative electron acceptor under hypoxic conditions^{34,47}. The

increased essentiality of the *csd* gene observed in the clinical MAC-PD strains seems to reflect the switching of the electron-transport system to less-energy-efficient respiration to support preferential hypoxic survival, including *in vivo*.

We tried to use inhibitors of phosphoenolpyruvate carboxykinase to confirm the increased essentiality of the *pckA* gene. However, we detected no positive effect because millimolar 3-mercaptopycolinic acid is required to show the effect of inhibition, even in eukaryotic cells, which have no cell wall⁴⁸. Moreover, 3-alkyl-1,8-dibenzylxanthine⁴⁹ also only effectively inhibits PckA in eukaryotic cells. However, PckA has been proposed as a drug target for Mtb⁵⁰, so our TnSeq data seem to be consistent with studies directed towards antimycobacterial drug discovery.

In summary, we have identified 131 universal essential and growth-defect-associated genes that covers the genomic diversity of *M. intracellulare* strains responsible for MAC-PD, which narrowed down the drug targets against MAC-PD. Furthermore, we have demonstrated a preferential adaptation to hypoxia in clinical NTM strains compared with the type strain in terms of the increased essentiality of genes required for hypoxic growth, such as those encoding phosphoenolpyruvate carboxykinase, the ESX-5 type VII secretion system, and cysteine desulfurase. These genes were shown to have a significant impact on infection *in vivo* by increasing their gene requirements. Our data highlight the importance of using clinical strains and hypoxic conditions in experimental research, and provide fundamental resources for developing novel drugs directed towards nontuberculous mycobacterial strains that cause severe clinical diseases.

Methods

Bacterial strains and phages used to prepare Tn mutants

A total of nine strains were used in this study, including seven clinical strains isolated from patients with MAC-PD in Japan (M.i.198, M.i.27, M018, M003, M019, M021, and M001)⁵¹, *M. intracellulare* genovar *paraintracellulare* type strain MOTT64 isolated from MAC-PD in South Korea⁵², and the *M. intracellulare* type strain ATCC13950 (Fig. 1). The Tn mutant libraries were constructed using glucose as a carbon source under aerobic conditions according to the transduction method used for ATCC13950⁹, except that a higher concentration of kanamycin (100 µg/ml) was used to select the transposon mutants than was used for ATCC13950 (25 µg/ml). In brief, a hundred milliliter of the bacteria were cultured aerobically in 7H9/ADC broth until mid-log phase (optical density at a wavelength of 600 nm [OD₆₀₀] of approximately 0.4–0.6). After harvesting the bacteria, 10 mL of high-titer mycobacteriophage phAE180⁵³ was transduced into the bacteria 1-day at 37°C for inserting Tn and plated on 7H10/OADC agar plates containing kanamycin. After 2 or 3 week-cultivation at 37°C aerobically, the colonies were harvested *en masse* and used as Tn mutant libraries. The kanamycin-resistant colonies ($> 1 \times 10^5$) were harvested and evenly resuspended in 7H9/ADC medium containing 20% glycerol, aliquoted, and stored at –80 °C until further use.

Next-generation sequencing and TnSeq analysis

DNA libraries were prepared as reported previously⁵⁴. The TnSeq libraries were sequenced with the NextSeq 500 System, 150-bp single-end run (Illumina, San Diego, CA, USA). The sequence reads were analyzed as previously described⁹. Gene essentiality was calculated with resampling and HMM analyses using TRANSIT¹¹. First, an HMM analysis was conducted to calculate the fluctuation of the number of Tn insertion reads individually for each strain. Next, a resampling analysis was conducted to calculate the difference of the number of Tn insertion reads between each clinical strain and ATCC13950 (Supplementary Table 6). By comparing the data obtained by HMM and resampling analyses, we summarized the difference of gene essentiality between the clinical MAC-PD strains and ATCC13950. To compare the gene essentiality of the clinical MAC-PD

strains and ATCC13950, the annotation for the clinical MAC-PD strains was adapted from that of ATCC13950 by adjusting the START and END coordinates of each open reading frame (ORF) in the clinical MAC-PD strains according to their alignment with the corresponding ORFs of ATCC13950. The function of TnSeq-hit genes was categorized according to the Kyoto Encyclopedia of Genes and Genomes (KEGG) database.

Comparison of growth curves under aerobic and hypoxic conditions

First, we precultured the bacteria in 7H9/ADC broth until the bacteria reached early log-phase (OD_{600} approximately 0.1–0.3). Next, the bacterial cultures were diluted with new 7H9/ADC broth (final OD_{600} 0.003) for the aerobic and hypoxic growth assay. Bacterial cells were grown statically in 96-well plates containing 250 μ l of 7H9/ADC broth in each well under aerobic conditions or 5% oxygen at 37 °C with humidification (APM-30D incubator, Astec, Tokyo, Japan). The culture was sampled every 1–3 days. At sampling, the culture was mixed by pipetting 50 times. An aliquot (4 μ l) of the culture and its diluents were streaked onto 7H10/OADC agar and the CFUs counted. The growth curve was fitted to the logistic 4P model with the JMP software (SAS Institute Inc., Cary, NC, USA) to calculate the inflection point (midpoint) and growth rate at midpoint in the log growth phase. Entry into the log phase was evaluated from the timing of the inflection point on the growth curve.

Animals

Female C57BL/6Jcl mice were purchased from CLEA Japan (Tokyo, Japan). Six-week-old mice were used for the infection experiment (Supplementary Fig. 1). All the mice were kept under specific-pathogen-free conditions in the animal facility of Niigata University Graduate School of Medicine, according to the institutional guidelines for animal experiments. The mice were housed (four per cage) under a 12-h/12-h light/dark cycle. Food and water were available *ad libitum*. Kanamycin was added to the drinking water to a final concentration of 100 μ g/ml to maintain the mice's resistance to the Tn mutant strains. Euthanasia was conducted with an intraperitoneal injection of a combined anesthetic (15 μ g of medetomidine, 80 μ g of midazolam, and 100 μ g of butorphanol), followed by cervical dislocation.

Mouse TnSeq experiment

The protocol was based on our previous mouse lung infection experiment⁵¹. Twenty mice each were used for the Tn mutant strains M.i.198 and M.i.27. The mice were injected intratracheally with 3.25×10^8 and 5.93×10^7 CFUs of the Tn mutant strains, respectively. The infected lungs were expected to be harvested from five mice at day 1, week 4, week 8, and week 16 after infection. However, the lungs were harvested from only four mice infected with the M.i.27 Tn mutant strains in week 8 because we experienced difficulties with sampling, and the lungs were harvested from only three, one, and two mice infected with M.i.198 Tn mutant strains in week 8, week 12, and week 16, respectively, because morbidity was high in the mice infected with M.i.198. The harvested lungs were homogenized in 4.5 ml of distilled water with the gentleMACS™ Tissue Dissociator (Miltenyi Biotec, Bergish Gladbach, Germany) and the homogenates were plated on Middlebrook 7H10–oleic acid ADC (7H10-OADC) agar plates and cultured for 3 weeks. The colonies were harvested *en masse* for TnSeq.

Gene set enrichment analysis

To investigate the overlap of the gene sets required for growth in mouse lungs with those required for hypoxic pellicle formation observed in ATCC13950 and those showing increased gene essentiality observed in the clinical MAC-PD strains, a statistical enrichment of gene sets was

performed by GSEA 4.3.2²⁰. Parameters set for GSEA were as follows: permutations = 1,000, permutation type: gene_set, enrichment statistic: weighted, metric for ranking genes: Signal2Noise, max size: 500, min size: 15.

Construction and growth assay of knockdown strains

Knockdown strains were constructed with the CRISPR-dCas9-mediated RNA interference system, as reported previously²⁷. Gene-specific 20-nt sequences were selected as protospacers for incorporation into the sgRNAs (Supplementary Table 13). pRH2502 (a vector expressing an inactivated version of the *cas9* DNA sequence) and pRH2521 (an sgRNA-expressing vector whose expression is regulated by the TetR-regulated smyc promoter $P_{myc1tetO}$) were introduced into the bacteria to obtain colonies of the knockdown strains. For selection, 25 µg/ml kanamycin and 50 µg/ml hygromycin were used for ATCC13950, and 100 µg/ml kanamycin and 50 µg/ml hygromycin were used for the clinical strains. The growth assay of the knockdown strains was performed in 7H9/ADC broth containing the above-mentioned concentrations of kanamycin and hygromycin under aerobic conditions at 37 °C with humidification (APM-30D incubator, Astec, Tokyo, Japan). During the assay, the cultures were supplemented with anhydrotetracycline (aTc) to a final concentration of 50 ng/ml (as a concentration that is not toxic to ATCC13950 and M.i.27) or 200 ng/ml (for the other clinical strains). aTc was added repeatedly every 48 h to maintain the induction of dCas9 and sgRNAs in experiments that extended beyond 48 h²⁷. An aliquot (4 µl) of the culture and its diluents were streaked onto 7H10/OADC agar plates containing kanamycin and hygromycin, and the CFUs were counted. Growth inhibition was evaluated as the ratio of the CFUs of knockdown strains compared with those of the vector control strains lacking the guide RNA sequence. To confirm the downregulation of gene expression in the knockdown strains, quantitative reverse transcription PCR (qRT-PCR) was performed. Bacteria were cultured in 10 ml of Middlebrook 7H9 medium (Difco Laboratories, Detroit, MI, USA) containing 0.2% glycerol, 0.1% Tween 80 (MP Biochemicals, Illkrich, France), and 10% Middlebrook albumin–dextrose–catalase (7H9/ADC/Tween 80), kanamycin, and hygromycin until mid-log phase (OD_{600} approximately 0.4–0.6). The cultures were supplemented with aTc every 48 h, as in the growth assay for the knockdown strains. One day after the second addition of aTc, RNA was extracted by bead-beating six times at 5000 rpm for 30 s with cooling (Micro Smash MS 100R Cell Disruptor; Tomy Seiko, Tokyo, Japan) and cDNA was synthesized with the Direct-zol™ RNA Miniprep (Zymo Research, Orange, CA, USA) and ReverTra Ace qPCR RT Master Mix with gDNA Remover (Toyobo, Osaka, Japan). qRT-PCR was performed with the CFX Connect Real-time PCR Detection System (Bio-Rad, Hercules, CA, USA), according to the manufacturer's instructions. The data were normalized to the expression of *sigA*.

Statistical analyses

Pearson's correlation test was performed between the fitness costs ($\log_2FC$ calculated by the resampling assay), efficiency of knockdown (assessed by qRT-PCR) and growth suppression in the knockdown strains. Statistical analyses of the growth curves were performed using a 4-parameter logistic model (logistic 4P model), Wilcoxon two-sample test, and a one-way analysis of variance (ANOVA) followed by Steel's multiple comparisons test using the JMP software program (SAS Institute Inc., Cary, NC, USA). Data are presented as the mean $\pm$ SD. Statistical significance was set as *P* values less than 0.05.

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Author contributions

Y.T. conceived, designed and performed the experiments. Y.O., A.N., and S.M. contributed reagents or materials. Y. Morishige., Y. Minato, and A.D.B. contributed analytical tools. Y. Morishige., Y. Minato, A.D.B., and S.M. provided critical suggestions throughout the preparation of the manuscript. Y.T. wrote the manuscript. All authors read and approved the final manuscript.

Availability of data and materials

The datasets used or analyzed in this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

All experimental protocols were approved by the Institutional Review Board of Niigata University (approval numbers: 2019-0020 for the use of human tissue samples and SA00506 for experiments on live vertebrates). All experiments on humans and human samples were conducted in accordance with relevant guidelines and regulations. Informed consent was obtained from all patients. All experiments on animals were conducted in accordance with the relevant guidelines and regulations, and the study complied with the ARRIVE guidelines.

Consent for publication

According to the protocols approved by the Institutional Review Board of Niigata University mentioned above, consent for publication was obtained from the patients enrolled in this study.

Competing interests

The authors declare that they have no competing interests.

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Reviewer #1 (Public Review):

Summary:

Tateishi et al. report a Tn-seq-based analysis of genetic requirements for growth and fitness in 8 clinical strains of *Mycobacterium intracellulare* (Mi), and compare the findings with a

type strain ATCC13950. The study finds a core set of 131 genes that are essential in all nine strains, and therefore are reasonably argued as potential drug targets. Multiple other genes required for fitness in clinical isolates have been found to be important for hypoxic growth in the type strain.

Strengths:

The study has generated a large volume of Tn-seq datasets of multiple clinical strains of Mi from multiple growth conditions, including from mouse lungs. The dataset can serve as an important resource for future studies on Mi, which despite being clinically significant remains a relatively understudied species of mycobacteria.

Weaknesses:

The paper lacks clarity in data presentation and organization. For example, some of the key data on cfu counts of clinical Mi strains in a mouse model can be presented along with the Tn-seq dataset in Figure 6, the visualization of which can be improved with volcano plots. etc. Improvement in data visualization is perhaps necessary throughout the paper.

The primary claim of the study that the clinical strains are better adapted for hypoxic growth is not well-supported by the data presented in Figure 7.

The title of the paper is misleading as the study doesn't provide any mechanistic aspect of hypoxic adaptation in Mi.

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Reviewer #2 (Public Review):

Summary:

In the study titled "Functional genomics reveals the mechanism of hypoxic adaptation in nontuberculous mycobacteria" by Tateishi et al., the authors have used TnSeq to identify the common essential and growth-defect-associated genes that represent the genomic diversity of clinical *M. intracellulare* strains in comparison to the reference type strain. By estimating the frequency of Tn insertion, the authors speculate that genes involved in gluconeogenesis, the type VII secretion system, and cysteine desulfurase are relatively critical in the clinical MAC-PD strains than in the type strain, both for the extracellular survival and in a mouse lung infection model.

Based on their analysis, the authors proposed to identify the mechanism of hypoxic adaptation in nontuberculous mycobacteria (NTM) which offer promising drug targets in the strains causing clinical *Mycobacterium avium*-intracellulare complex pulmonary disease (MAC-PD).

Strengths:

A major strength of the manuscript is the performance of the exhaustive set of TnSeq experiments with multiple strains of *M. intracellulare* during in vitro growth and animal infection.

Weaknesses:

(1) The study suffers from the authors' preconceived bias toward a small subset of genes involved in hypoxic pellicle formation in ATCC13950.

(2) An important set of data with the ATCC13950 reference strain is missing in the mouse infection study. In the absence of this, it is difficult to establish whether the identified genes are critical for infection/intracellular proliferation, specifically in the clinical isolates that are relatively more adapted for hypoxia.

(3) Statistical enrichment analysis of gene sets by GSEA wrongly involves genes required for hypoxic pellicle formation in ATCC13950 together with the gene sets found essential in the clinical MAC-PD strains, to claim that a significant % of genes belong to hypoxia-adaptation pathways. It could be factually incorrect because a majority of these might overlap with those found critical for the in vitro survival of MAC-PD strains (and may not be related to hypoxia).

(4) Validation of mouse infection experiments with individual mutants is missing.

(5) Phenotypes with TnSeq and CRISPRi-based KD exhibit poor correlation with misleading justifications by the authors.

In summary, this study is unable to provide mechanistic insights into why and how different MAC-PD mutant strains exhibit differential survival (in vitro and in animals) and adaptation to hypoxia. It remains to understand why the clinical strains show better adaptation to hypoxia and what is the impact of other stresses on their growth rates.

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Reviewer #3 (Public Review):

Summary:

The study by Tateishi et al. utilized TnSeq in nine genetically diverse *M. intracellulare* strains, identifying 131 common essential and growth-defect-associated genes across those strains, which could serve as potential drug targets. The authors also provided an overview of the differences in gene essentiality required for hypoxic growth between the reference strain and the clinical strains. Furthermore, they validated the universal and accessory/strain-dependent essential genes by knocking down their expression using CRISPRi technique. Overall, this study offers a comprehensive assessment of gene requirements in different clinical strains of *M. intracellulare*.

(1) The rationale for using ATCC13950 versus clinical strains needs to be clarified. The reference strain ATCC13950 was obtained from the abdominal lymph node of a patient around 10 years ago and is therefore considered a clinical strain that has undergone passages in vitro. How many mutations have accumulated during these in vitro passages? Are these mutations significant enough to cause the behavior of ATCC13950 to differ from other recently sampled clinical strains? From the phylogenetic tree, ATCC13950 is located between M018 and M.i.27. Did the authors observe a similarity in gene essentiality between ATCC13950 and its neighbor strains? What is the key feature that separates ATCC13950 from these clinical strains? The authors should provide a strong rationale for how to interpret the results of this comparison in a clinical or biological context.

(2) Regarding the 'nine representative strains of *M. intracellulare* with diverse genotypes in this study,' how were these nine strains selected? To what extent do they represent the genetic diversity of the *M. intracellulare* population? A phylogenetic tree illustrating the global genetic diversity of the *M. intracellulare* population, with these strains marked on it, would be important to demonstrate their genetic representativeness.

(3) The authors observed a considerable amount of differential gene requirements in clinical strains. However, the genetic underpinning underlying the differential requirement of genes

in clinical strains was not investigated or discussed. Because *M. intracellulare* has a huge number of accessory genes, the authors should at least check whether the differential requirement could be explained by the existence of a second copy of functional analogous genes or duplications.

(4) Growth in aerobic and hypoxic conditions: The authors concluded that clinical strains are better adapted to hypoxia, as reflected by their earlier entry into the log phase. They presented the 'Time at midpoint' and 'Growth rate at midpoint.' However, after reviewing the growth curves, I noticed that ATCC13950 had a longer lag phase compared to other strains under hypoxic conditions, and its phylogenetic neighbor M018 also had a longer lag phase. Hence, I do not believe a conclusion can be drawn that clinical strains are better adapted to hypoxia, as this behavior could be specific to a particular clade. It's also possible that the ATCC13950 strain has adapted to aerobic growth. I would suggest that the authors include growth curves in the main figures. The difference in 'Time at midpoint' could be attributed to several factors, and visualizing the growth curves would provide additional context and clarity.

(5) Lack of statistical statement: The authors emphasized the role of pellicle-formation-associated genes in strain-dependent essential and accessory essential genes. Additionally, the authors observed that 10% of the genes required for mouse infection are also required for hypoxic pellicle formation. However, these are merely descriptive statements. There is no enrichment analysis to justify whether pellicle-formation-associated genes are significantly enriched in these groups.

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